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Parameter Identification of PMSM Based on Improved Secretary Bird Optimization Algorithm

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Abstract

Permanent magnet synchronous motors (PMSMs) serve as the critical propulsion component in electric aircraft systems, where accurate parameter identification is essential for optimal performance. To address the limitations of conventional methods in balancing precision and computational efficiency, this study develops an improved Secretary Bird Optimization Algorithm (ISBOA) featuring multiple algorithmic enhancements. The proposed approach incorporates Tent chaotic mapping for population initialization, stochastic differential mutation for solution diversity preservation, random interaction transfer for local optima avoidance, and adaptive step-size adjustment for balanced search performance. Successfully implemented on a TMS320F28335 DSP-based control platform, ISBOA demonstrates superior experimental performance with parameter identification errors below 2.6% and significantly faster convergence compared to traditional methods. This work provides both algorithmic advances in optimization techniques and a practical solution for real-time motor control in electric aviation applications.

Keywords: permanent magnet synchronous motor, electric aircraft, parameter identification, secretary bird optimization algorithm

1 Introduction

Electric aircraft rely on electricity as their power source, eliminating exhaust emissions during operation [1–3]. Electric aircraft reduce emissions and pollution while offering cost-efficiency and operational advantages, representing the future of sustainable aviation [4, 5]. To achieve optimal performance, electric aircraft must feature propulsion systems that are lightweight, compact, and have high power density [6–8]. Permanent Magnet Synchronous Motors (PMSMs) use rare-earth magnets to provide high power in a compact, lightweight design, making them ideal for electric aircraft [9, 10]. The precise control of a motor is inherently tied to its motor parameters, such as dq-axis inductance, stator resistance, and magnetic chain characteristics [11]. Electric aircraft face changing environmental conditions—such as high-altitude heat or shaded cooling—that cause motor temperature fluctuations, altering key parameters (dq-axis inductance, stator resistance, flux linkage)[9]. To ensure precise control, accurate and efficient parameter identification methods are essential [12–14]. Such a method would enable real-time monitoring and dynamic adjustment of motor performance, ensuring optimal efficiency despite external variations encountered during flight [15].

Various strategies have been developed to improve the accuracy of PMSM parameter identification. These strategies can generally be categorized into traditional methods and those based on artificial intelligence optimization algorithms [16]. Traditional methods include the Model Reference Adaptive System (MRAS) [17, 18], Extended Kalman Filter (EKF) [19–21], and Recursive Least Squares (RLS) [22–24]. In [17], the Improved MRAS method enables online identification of PMSM stator inductance by combining Adjustable Model Predictive Control (AMPC) with a fuzzy-based adaptation law, improving accuracy and disturbance rejection. However, its linearization process may reduce accuracy in some cases. In [19], EKF is used to identify rotor flux and inductance in lunar rovers, offering strong anti-interference capabilities and real-time parameter updates. However, the linearization of the nonlinear system can introduce errors, which limits its precision in certain situations. In [21], RLS achieves efficient real-time parameter estimation for PMSMs in both steady and transient states, but suffers from potential performance degradation due to data saturation.

Given the limitations of traditional methods, researchers have turned to optimization algorithms for PMSM parameter estimation, yielding promising results in overcoming the challenges of nonlinearity and time variance. In [25], an enhanced GA improves Permanent Magnet Synchronous Wind Generator (PMSWG) parameter identification’s optimization, convergence and accuracy, but lacks comparison with other optimization algorithms. The MRAS-SAPSO method in [26] identifies electrical parameters via MRAS before optimizing mechanical parameters using SAPSO. While achieving higher accuracy, it requires substantial computation and converges slowly. The Chaotic Gaussian-Cauchy RAO algorithm (CGCRAO) in [27] combines Gaussian and Cauchy functions to avoid local optima, employing Tent chaotic mapping for population initialization to boost diversity. While easily implementable in embedded systems with minimal configuration, its inductance optimization accuracy needs enhancement.

The Secretary Bird Optimization Algorithm (SBOA) [28] was developed in 2024 to solve optimization problems by mimicking secretary birds’ hunting behaviors. While

SBOA shows fast convergence and robustness, it suffers from local optima issues when identifying PMSM parameters. To address these limitations, an improved ISBOA algorithm was created. ISBOA enhances SBOA through tent chaotic mapping for better initial population distribution, stochastic differential variation for increased diversity, and adaptive step-size control for refined optimization. Experimental results confirm ISBOA's superior performance over PSO, E.WOA and SBOA in both convergence speed and parameter identification accuracy for PMSM applications.

2 Pmsm mathematical model

PMSMs are categorized into built-in and surface-mounted types based on their structures [29]. This paper focuses on the surface-mounted type, which is a strongly coupled, nonlinear, and complex system [30]. To simplify the analysis, the effects of iron loss and eddy current loss are ignored. The PMSM voltage equation in the ABC coordinate system is presented in Eq. (1):

$$\begin{cases} u_a = Ri_a + \frac{d\psi_a}{dt} \\ u_b = Ri_b + \frac{d\psi_b}{dt} \\ u_c = Ri_c + \frac{d\psi_c}{dt} \end{cases} \quad (1)$$

where $u_a, u_b, u_c, i_a, i_b, i_c, \psi_a, \psi_b$, and ψ_c are the voltage, current and flux linkage in abc coordinates respectively. The stator voltage equation for PMSM in the dq coordinate system is shown in Eq. (2):

$$\begin{cases} u_d = Ri_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \\ u_q = Ri_q + L_q \frac{di_q}{dt} - \omega_e L_d i_d + \psi_f \omega_e \end{cases} \quad (2)$$

where u_d, u_q, i_d, i_q are the dq-axis stator voltage and current respectively, L_d, L_q are the dq-axis inductance, ω_e is the rotor electrical angular velocity, and ψ_f is the permanent magnet flux linkage.

In the steady-state case where $\frac{di_d}{dt} = 0$ and $\frac{di_q}{dt} = 0$, Eq. (2) simplifies to Eq. (3):

$$\begin{cases} u_d = R_s i_d - \omega_e L_q i_q \\ u_q = R_s i_q + \omega_e L_d i_d + \omega_e \psi_f \end{cases} \quad (3)$$

3 ISBOA algorithm

3.1 SOBA

SBOA is a new metaheuristic algorithm inspired by secretary birds' dual survival strategies: exploring (snake hunting) for global search and exploiting (predator evasion) for local refinement, dynamically balancing these behaviors to solve optimization problems. The basic principle is described as shown below:

1) Set and initialize the parameters, and randomly generate N individuals in the search space as the initial population for the computation. As shown in Eq. (4):

$$X_{i,j} = lb_j + r \times (ub_j - lb_j), i \in [1, N], j \in [1, Dim] \quad (4)$$

where $X_{i,j}$ is the location of the i^{th} secretary bird, lb_j and ub_j are the lower and upper bounds of the j^{th} dimension respectively, and r is a random number in $[0, 1]$.

2) The exploratory phase of the secretary bird consists of three phases: searching for prey, devouring prey, and attacking prey, and the entire hunting process is divided into three equal periods for mathematical modeling respectively. The process is mathematically described by Eq. (5), with the location update governed by Eq. (6).

$$\begin{cases} \text{While } t < \frac{1}{3}T, & x_{i,j}^{\text{new}, P1} = x_{i,j} + (x_{\text{random}_1} - x_{\text{random}_2}) \times r \\ \text{While } \frac{1}{3}T < t < \frac{2}{3}T, & x_{i,j}^{\text{new}, P1} = x_{\text{best}} + \exp((t/T)^4) \times (RB - 0.5) \times (x_{\text{random}} - x_{i,j}) \\ \text{While } \frac{2}{3}T < t, & x_{i,j}^{\text{new}, P1} = x_{\text{best}} + ((1 - t/T)^{2 \times t/T}) \times x_{i,j} \times RL \end{cases} \quad (5)$$

$$X_i = \begin{cases} X_i^{\text{new}, P1}, & \text{if } F_i^{\text{new}, P1} < F_i \\ X_i, & \text{else} \end{cases} \quad (6)$$

where t denotes the current iteration number, T represents the maximum number of iterations, $x_{i,j}^{\text{new}, P1}$ is the updated position of the i^{th} secretary bird. The positions x_{random_1} , x_{random_2} correspond to two randomly selected individuals from the population. Here, F_i indicates the fitness value of the i^{th} secretary bird, while $F_i^{\text{new}, P1}$ corresponds to the recalculated fitness value after the position update, and the superscript P1 denotes the exploratory phase. x_{best} is the current location of the optimal individual. Brownian motion (RB) is incorporated as defined in Eq. (7), and the Lévy flight function (RL) is expressed by Eq. (8).

$$RB = \text{randn}(1, \text{Dim}) \quad (7)$$

$$\begin{cases} RL = 0.5 \times \text{levy}(\text{Dim}) \\ \text{Levy}(D) = s \times \frac{\mu \times \sigma}{|v|^{1/\eta}} \end{cases} \quad (8)$$

where $s = 0.01$ and $\eta = 1.5$ are fixed constants, μ and v are random numbers $\in [0, 1]$. The equation for σ is shown in Eq. (9):

$$\sigma = \left(\frac{\Gamma(1 + \eta) \times \sin(\frac{\pi\eta}{2})}{\Gamma(\frac{1+\eta}{2}) \times \eta \times 2^{\frac{\eta-1}{2}}} \right)^{1/\eta} \quad (9)$$

3) Two escape strategies are employed in the development phase to update the secretary bird's position, as shown in Eq. (10) and Eq. (11):

$$x_{i,j}^{\text{new}, P2} = \begin{cases} C_1 : x_{\text{best}} + (2RB - 1) \left(1 - \frac{t}{T}\right)^2 x_{i,j}, & \text{if } \text{rand} < 0.5 \\ C_2 : x_{i,j} + R_2(x_{\text{random}} - Kx_{i,j}), & \text{otherwise} \end{cases} \quad (10)$$

$$X_i = \begin{cases} X_i^{\text{new}, P2}, & \text{if } F_i^{\text{new}, P2} < F_i \\ X_i, & \text{otherwise} \end{cases} \quad (11)$$

where R_2 is a random $1 \times \text{Dim}$ vector from a standard normal distribution, K is a random integer from $[1, 2]$, and the superscript P2 denotes the exploitation phase.

3.2 ISOBA

The SBOA is prone to problems such as low convergence accuracy and insufficient global exploration capability. These shortcomings lead to rapid clustering of the population in the late iteration, resulting in a loss of diversity and an increased probability of falling into a local optimum. In this study, we propose that ISBOA optimizes the aforementioned problem using four strategies.

3.2.1 Tent chaos mapping

The initial population distribution significantly impacts optimization algorithm performance. While SBOA uses random initialization that may result in uneven distribution and reduced diversity, we employ Tent chaotic mapping to generate more uniform initial populations. As defined in Eq. (12), this approach enhances population diversity and improves convergence speed.

$$Y_{i+1} = \begin{cases} Y_i/\alpha, & Y_i \in [0, \alpha) \\ (1 - Y_i)/(1 - \alpha), & Y_i \in [\alpha, 1) \end{cases} \quad (12)$$

According to the Tent chaotic mapping the initialized population can be changed to Eq. (13):

$$X_{i,j} = lb_j + Y_i \times (ub_j - lb_j), \quad i \in [1, N], \quad j \in [1, Dim] \quad (13)$$

where Y_i is a chaotic sequence in the interval $[0, 1]$ and $\alpha = 0.6$ is chosen in this paper.

Fig. 1 shows the uniform distribution of Tent chaotic mapping over $[0, 1]$ in 5000 iterations, ensuring population diversity for better optimization.

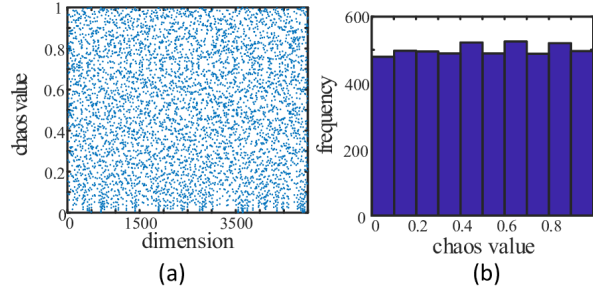


Fig. 1 Tent Chaotic map distribution. **a** Particle profile. **b** Frequency map.

3.2.2 Stochastic differential variation strategy

In the predation phase, the SBOA updates positions based on the current best solution. However, as iterations progress, the population tends to converge around local optima, reducing diversity and trapping the algorithm in local optima, limiting its global search capability. To address this, the ISBOA introduces a stochastic differential variation

strategy. This strategy adds randomness to position updates, enhancing diversity, preventing premature convergence, and improving global exploration. The strategy helps escape local optima while maintaining local search precision, thereby improving optimization efficiency, as shown in Eq. (14):

$$\text{While } \frac{1}{3}T < t < \frac{2}{3}T, x_{i,j}^{\text{newP1}} = r_1 \times (x_{\text{best}} - x_{i,j}) + r_2 \times (x_{\text{random}} - x_{i,j}) \quad (14)$$

where x_{random} is the position of a randomly selected individual from the population, and r_1, r_2 are random values in the range $[0, 1]$.

3.2.3 Stochastic interaction migration strategy

After completing the phases of prey detection, predation, and attack, the SBOA searches for new prey locations based on previous hunting positions or those of other birds. The ISBOA introduces a stochastic interaction migration strategy, where the positions of other birds are used as interaction factors, and one is randomly selected for migration. This expands the search space and increases predation success. Unlike traditional methods, this strategy avoids over-reliance on a single individual, preventing premature convergence. By enhancing diversity, it improves global exploration and optimization efficiency, as shown in Eq. (15):

$$x_{i,j}^{\text{new,P1}} = \begin{cases} x_{i,j} + r_1 \times (x_{\text{random}_1} - I_{i,j} \times x_{i,j}), & \text{if } F_i^{\text{new,P1}} < F_i \\ x_{i,j} + r_1 \times (x_{i,j} - x_{\text{random}_1}), & \text{otherwise} \end{cases} \quad (15)$$

3.2.4 Adaptive step size policy and adaptive minimum step size strategy

ISBOA overcomes SBOA's limitations by introducing adaptive step size strategies Eq. (16). The algorithm dynamically adjusts step sizes using a linearly decreasing factor α_0 and adaptive minimum step size L_o . Early iterations employ larger steps for global exploration, while later iterations use smaller steps for precise local search, effectively balancing exploration and exploitation.

$$\begin{cases} a_0 = \cos \left[\frac{\pi}{3} \times \left(1 + \frac{t}{T} \right) \right] \\ L_o = \frac{lb}{t} \end{cases} \quad (16)$$

where lb is the lower bound of the search range.

The position is updated based on the escape strategy, incorporating adaptive step size and minimum step size strategies, as shown in Eq. (17):

$$x_{i,j}^{\text{new,P2}} = \begin{cases} C1 : x_{i,j} + \alpha_0 \times \left(1 - \frac{t}{T} \right) \times (2 \times r_1) \times x_{i,j}, & \text{if } \text{rand} < r_i \\ C2 : x_{i,j} + \alpha_0 \times (x_{\text{random}} - K \times x_{i,j}) + L_o, & \text{otherwise} \end{cases} \quad (17)$$

According to the modification of the above policy, then the flowchart of ISBOA is shown in Fig. 2.

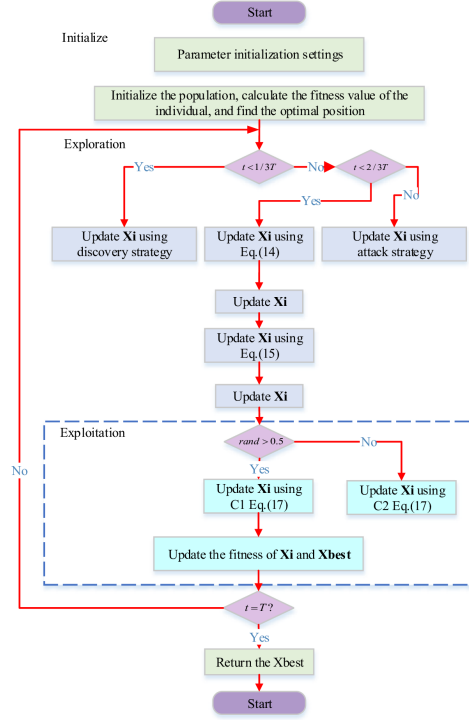


Fig. 2 Flowchart of ISBOA algorithm

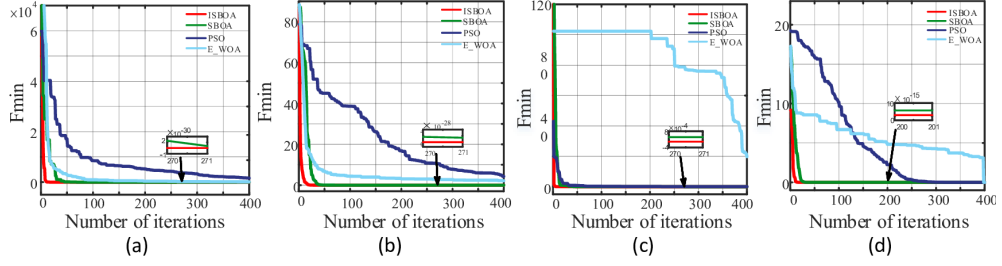
3.3 Algorithm performance testing

To determine if the modified algorithm demonstrates superior performance in optimization-seeking tasks, four representative benchmark functions were selected for evaluation within this study. SBOA, ISBOA, PSO, and E_WOA are introduced to compare the optimization-seeking effect. The specific information on the functions is shown in Table 1. Where f_1 , f_2 , and f_3 are single-peaked functions (with one and only one optimal value) used to detect the local exploitation of the algorithm. f_4 is a multi-peak function, characterized by multiple local minima within the specified interval, and serves to evaluate the algorithm's global exploration capability. The number of iterations of the above algorithm is set to 400; the population size is 50.

From Figs. 3(a)-(d), ISBOA achieves the fastest convergence to 0, typically attaining full convergence within 15 generations, while demonstrating the highest solution accuracy. SBOA and E_WOA show moderate convergence, completing within 30-50 generations, while PSO converges much slower, often failing to reach 0 even after 400 generations. Overall, ISBOA outperforms SBOA, E_WOA, and PSO in convergence speed, accuracy, and stability, with smoother convergence curves.

Table 1 Information on the 4 standard functions

Function	Definition	Dimension	Range
$f_1(x)$	$\sum_{i=1}^n \left(\sum_{j=1}^n x_j \right)^2$	30	$[-100, 100]$
$f_2(x)$	$\max x_i $	30	$[-100, 100]$
$f_3(x)$	$\sum_{i=1}^n i \cdot x_i^4 + \text{rand}$	30	$[-1.28, 1.28]$
$f_4(x)$	$-20 \exp \left(-0.2 \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} \right) - \exp \left(\frac{1}{n} \sum_{i=1}^n \cos(2\pi x_i) \right) + 20 + e$	30	$[-32, 32]$

**Fig. 3** Test function search results graph. **a** f_1 test results. **b** f_2 test results. **c** f_3 test results. **d** f_4 test results

4 ISBOA-Based PMSM parameter identification

4.1 PMSM parameter identification principle

In this paper, we take the table-posted PMSM as the identification object, whose structure makes $L_d = L_q = L_s$ [20], so the PMSM needs to identify at least three motor parameters $\{R_s, L_s, \psi_f\}$. Specifically, Eq. (3) represents a system with a rank deficiency of 2, making it unsolvable for three unknown quantities. In this paper, based on the vector control with $i_d = 0$, by adding the case where $i_d \neq 0$ during steady-state motor operation, an additional set of equations can be generated to form Eq. (18), which solves the problem of rank deficiency [31, 32]. The PMSM parameter identification discrete equations are shown in Eq. (18):

$$\begin{cases} \hat{u}_{d0}(k) = -\hat{L}_q \omega_{e0}(k) i_{q0}(k) \\ \hat{u}_{q0}(k) = \hat{R}_s i_{q0}(k) + \hat{\psi}_f \omega_{e0}(k) \\ \hat{u}_d(k) = \hat{R}_s i_d(k) - \hat{L}_q \omega_e(k) i_q(k) \\ \hat{u}_q(k) = \hat{R}_s i_q(k) + \hat{L}_d \omega_e(k) i_d(k) + \hat{\psi}_f \omega_e(k) \end{cases} \quad (18)$$

Variables marked with $\hat{\cdot}$ are estimated values based on the model analysis, while unmarked variables are actual measurements.

In the process of motor parameter identification, the differences between the identification model values and the actual measured values are first compared, and then the ISBOA algorithm is used to adjust the parameters to be determined in the identification model. The objective of this process is to reduce the value of the fitness function in order to identify the optimal motor parameter configuration. The fitness function is shown in Eq. (19):

$$\begin{aligned} fitness = & \omega_1 \times |u_{d0}(k) - \hat{u}_{d0}(k)| + \omega_2 \times |u_d(k) - \hat{u}_d(k)| \\ & + \omega_3 \times |u_q(k) - \hat{u}_q(k)| + \omega_4 \times |u_{q0}(k) - \hat{u}_{q0}(k)| \end{aligned} \quad (19)$$

where $\omega_1, \omega_2, \omega_3, \omega_4$ are the weight coefficients of the adaptive function; u_{d0} and u_{q0} are dq -axis voltage measurements under the $i_d = 0$ control; $\hat{u}_{d0}, \hat{u}_{q0}$ are the estimates obtained by collecting data through the algorithm under the control of $i_d = 0$; u_d, u_q are dq -axis voltage measurements obtained when the motor injected negative sequence current i_d is not equal to 0; $\hat{u}_d, \hat{u}_q$ are estimates obtained by collecting data through an algorithm when the injected negative sequence current i_d is not equal to zero.

4.2 ISBOA-based PMSM parameter identification steps

The flow of the ISBOA is as follows:

Step 1: Run the PMSM and collect motor data for both $i_d = 0$ and $i_d \neq 0$, including electrical angular velocity, dq -axis voltage, and current.

Step 2: Tent chaotic mapping is employed to initialize the population, ensuring a more uniform distribution of the initial population. The improved initial population positions are obtained by substituting values based on the chaotic sequence generated by Eq. (13).

Step 3: Evaluate individual fitness and find the minimum fitness value f_{best} with its corresponding position Best_P.

Step 4: At stage $t < \frac{1}{3}T$, Eq. (5) and Eq. (6) are executed to perform the position update. At stage $\frac{1}{3}T < t < \frac{2}{3}T$, a stochastic differential variation strategy is applied to enhance the diversity of the secretary bird population. Eq. (14) and Eq. (6) are executed to perform the position update. At stage $\frac{2}{3}T < t$, Eq. (5) and Eq. (6) are executed to perform the position update.

Step 5: Implement a stochastic interaction migration strategy to simulate the authentic behaviors of secretary birds, preventing over-reliance on any individual during population updates. This enhances global exploration ability. Eq. (15) is executed to perform the position update.

Step 6: Introduce an adaptive step-size policy with an adaptive minimum step-size policy in the escape phase and Eq. (17) is executed to update the position.

Step 7: If the fitness value of the new position is better than the previous one, replace the old position with the new one. If the maximum number of iterations has not been reached, repeat steps 3 through 6. Once the maximum number of iterations is reached, proceed to step 8.

Step 8: Take the optimal fitness value from the iteration T -steps and compare it with the optimal individual position i.e. $\{R_s, L_s, \psi_f\}$.

5 Simulation and experimentation

5.1 Experimental parameter setting

A PMSM identification model based on the ISBOA has been constructed in Matlab. The motor employed in this study is a surface-mounted PMSM, and its system block diagram is illustrated in Fig. 4. The motor adopts a vector control strategy, and the input signals of the ISBOA are $i_d, i_q, u_d, u_q, \omega_r$, the specific parameters in the experiment are shown in Table 2.

Additional operational parameters of the motor include a main voltage of 150V, a switching frequency of 10kHz. The total duration of the simulation is 1 second. When PSO, E.WOA, SBOA, and ISBOA are incorporated into the simulation for comparative experiments, the population sizes of the four algorithms are 50, and the maximum number of iterations is 100.

Table 2 Specific parameters of the motor in the model

Electric Machine Parameter	Numerical Value
Number of pole pairs	4
Stator resistance R_s (Ω)	2.35
Q-axis inductance L_q (H)	0.0265
D-axis inductance L_d (H)	0.0265
Flux linkage ψ_f (Wb)	0.0101
Nominal torque T (N·m)	12.7
Rated speed ω (rpm)	3000

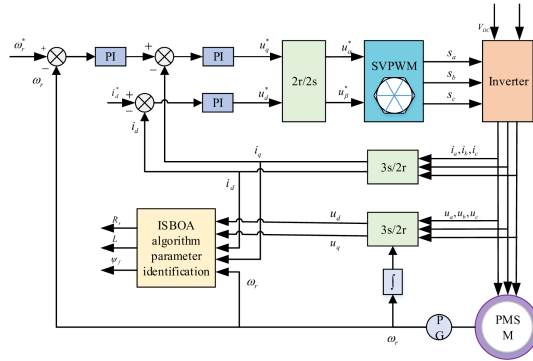


Fig. 4 Diagram of PMSM identification framework based on ISBOA

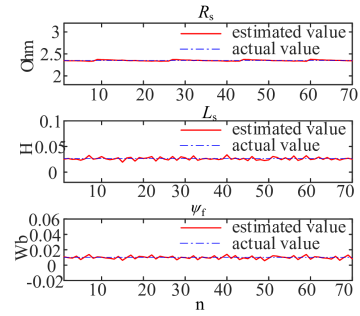


Fig. 5 Simulation Results of Parameter Identification for 70 Data Sets

5.2 Simulation verification

During the simulation, the d -axis current is alternated every 0.1 seconds between 0 A and -1 A. Since the data acquisition duration is short, it is assumed that the motor temperature remains constant throughout the process. The steady-state operating data of the motor are saved to the MATLAB workspace, with a total of 70 data sets collected over a specific time period. Each data set includes measurements of the dq -axis voltages, currents, and electrical angular velocity. These measurements are used to construct the fitness function for motor parameter identification. The fitness function is then solved using the ISBOA algorithm, with an initial population size of 50 and 200 iterations. The identification results from the 70 data sets are shown in Fig. 5.

As shown in Fig. 5 and Table 3, the three identified parameters effectively converge to values close to the real values. The errors presented in the table do not exceed 1%, indicating that the ISBOA algorithm achieves high solution accuracy in reverse parameter identification. This also highlights the superiority and strong solving capability of the algorithm in the parameter identification process.

Table 3 ISBOA parameter identification results

Evaluation Index	R_s (Ω)	L_s (H)	ψ_f (Wb)
Real Value	2.35	0.0265	0.0101
Maximum	2.3771	0.0338	0.0142
Minimum	2.3324	0.0195	0.0058
Average	2.3506	0.0263	0.01
Average Error (%)	0.026	0.075	0.99

The comparative error analysis for 70 data sets is shown in Table 4.

Table 4 Comparison of Errors in Four Parameter Identification Algorithms

Algorithm and Error	Parameters		
	R_s (Ω)	L_s (H)	ψ_f (Wb)
Real Value	2.35	0.0265	0.0101
ISBOA	2.3506	0.0263	0.0100
ISBOA Error (%)	0.026	0.075	0.99
SBOA	2.361	0.0259	0.0098
SBOA Error (%)	0.168	-2.26	-2.97
PSO	2.3507	0.0264	0.0100
PSO Error (%)	0.03	-0.377	0.99
E_WOA	2.3567	0.0237	0.0092
E_WOA Error (%)	0.29	-10.6	-8.91

From Table 4, ISBOA achieves the highest accuracy, with minimal errors for R_s (0.026%) and L_s (0.075%). While its error for ψ_f is slightly higher (0.99%), it remains

lower than other methods. In contrast, SBOA shows larger deviations, particularly for L_s (-2.26%) and ψ_f (-2.97%). PSO performs well for R_s (0.03%) and ψ_f (0.99%) but has a slight error in L_s (-0.377%). E_WOA performs the worst, with significant errors in L_s (-10.6%) and ψ_f (-8.91%). Overall, ISBOA outperforms the others, maintaining average errors below 1% for all parameters.

5.3 Experimental verification

To validate the proposed method, experiments were conducted using the Texas Instruments (TI) motor control platform TMDSHVMTRPFCKIT. This platform, based on the TMS320F28335 processor, provides robust real-time control capabilities, making it well-suited for motor control and parameter identification tasks. In the experiment, the frequency and sampling frequency were set to 10 kHz. The motor used was the EMJ-04APB22 model.



Fig. 6 Motor control platform

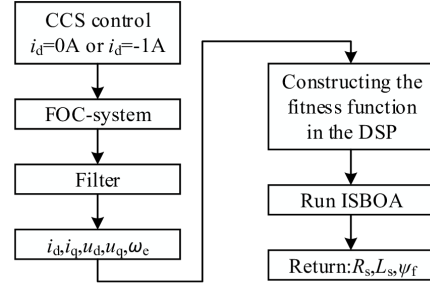


Fig. 7 Parameter identification flow chart

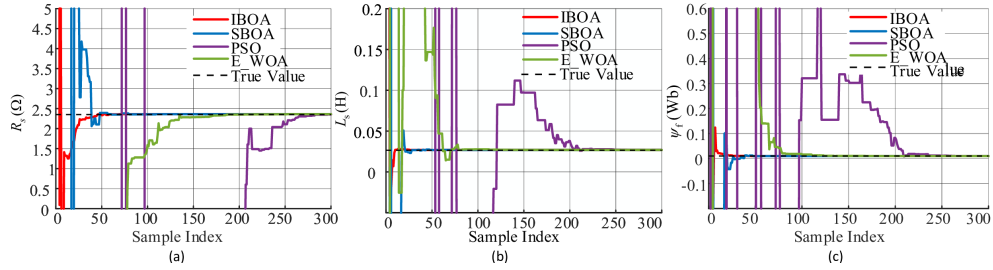


Fig. 8 Comparison of recognition results of four algorithms

The experimental procedure was as follows: First, the motor control program was developed and uploaded using the CCS (Code Composer Studio) environment to ensure stable motor operation and real-time data acquisition. The program, provided by TI, is the example file `HVMotorCtrl+PfcKit_v2.1\HVPM_Sensored_Servo`. During

Table 5 Comparison of Identification Results from Four Algorithms Using a Single Dataset

Algorithm and Error	Parameters		
	R_s (Ω)	L_s (H)	ψ_f (Wb)
Real Value	2.35	0.0265	0.0101
ISBOA	2.315	0.0262	0.01023
ISBOA Error (%)	-1.49	-1.13	1.29
SBOA	2.257	0.0272	0.0096
SBOA Error (%)	-3.96	2.64	-4.95
PSO	2.398	0.0273	0.0105
PSO Error (%)	2.04	3.02	3.96
E_WOA	2.411	0.0254	0.0093
E_WOA Error (%)	2.60	-4.15	-7.92

the experimental process, the system first injects $i_d = 0$ A and $i_d = -1$ A d -axis currents, continuously monitoring and collecting data on parameters such as voltage, current, and speed. Based on the PMSM parameter identification principles outlined in Section 4.1, the collected data are utilized to construct the fitness function. Finally, the ISBOA algorithm is executed within the control chip to identify the electrical parameters. Fig. 6 illustrates the actual motor control platform, while Fig. 7 presents the specific process flow of parameter identification within the control chip.

A specific data set was selected for identification using the ISBOA, SBOA, PSO, and E_WOA algorithms, with the results shown in Fig. 8 and Table 5. The population size was set to 100, the number of iterations to 300, and the parameter identification range was set to $[0, 5]$. In the PSO algorithm, the learning factor was set to 1.4.

To better illustrate the identification results, the vertical axes in Figs. 8(a)–(c) are set to appropriate ranges. ISBOA demonstrates fast and stable convergence: in Fig. 8(a), it approaches the true R_s by the 45th generation; in Fig. 8(b), it identifies the L_s by the 20th generation; and in Fig. 8(c), it converges to the true ψ_f by the 23rd generation. In contrast, E_WOA, SBOA, and PSO exhibit slower convergence and lower stability. For example, in Fig. 8(a), E_WOA converges by the 130th generation, SBOA by the 50th, and PSO by the 270th. Similar trends are seen in Figs. 8(b) and (c). As shown in Table 5, ISBOA achieves the smallest computational error ($\pm 2\%$), closely matching the simulated parameter identification results.

Table 6 ISBOA parameter identification result table

Evaluation Index	R_s (Ω)	L_s (H)	ψ_f (Wb)
Maximum	2.4902	0.0259	0.0112
Minimum	2.3232	0.0256	0.0092
Average	2.382	0.0258	0.01025
Average Error (%)	1.36	-2.6	1.49

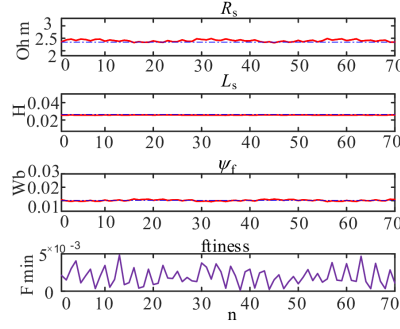


Fig. 9 Experimental Results of Parameter Identification for 70 Data Sets

In summary, ISBOA outperforms E_WOA, SBOA, and PSO in terms of faster convergence, smaller error, and better stability.

Seventy sets of motor operation data were collected and identified within the control chip, with the results shown in Fig. 9 and Table 6. The red line in the figure represents the estimated values from the ISBOA algorithm, the blue dashed line represents the actual values, and the purple line represents the value of the fitness function.

As shown in Fig. 9, after identification using 70 sets of parameters, the estimated values of the three parameters converge close to the real values, and the value of the fitness function is within 5×10^{-3} , demonstrating high convergence accuracy. According to Table 6, the error for the stator resistance R_s is 1.36%, the error for the inductance is -2.6% , and the error for the flux linkage is 1.49%, indicating that the identification method used has good accuracy and reliability.

6 Conclusion

This paper proposes ISBOA for motor parameter identification. The algorithm introduces four key enhancements to the standard SBOA: (1) Tent chaotic mapping for population initialization, (2) random differential mutation to enhance population diversity, (3) an interactive migration strategy to prevent premature convergence, and (4) adaptive step size adjustment to optimize search performance. Experimental results demonstrate that ISBOA outperforms comparative algorithms in convergence speed, stability, and accuracy, achieving simulation errors below 1% and physical experiment errors less than 2.6%. The proposed algorithm has been successfully deployed on embedded control chips, validating its practical value in real-time control systems and providing an efficient and reliable solution for motor parameter identification.

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Declarations

Competing interests . The authors declare no competing interests.